



ರೈ ತಾಂತ್ರಿಕ ವಿಶ್ವವಿದ್ಯಾಲಯ
Rai Technology University
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DEEP LEARNING

Assignment No: 2

MNIST Assignment

Subject Code: 0CS351T

Submitted By:

Student Name: Darshan L Siddu

University Enrolment Number: RTU24101CS027

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Submitted To:

Faculty Name: Mr Satyam Sir

Department of Computer Science & Engineering College Of
Engineering

Rai Technology University

Introduction

Deep learning is a branch of machine learning that uses neural networks to learn patterns from data. In this project, a neural network is used to recognize handwritten digits from 0 to 9.

The MNIST data set is used for training and testing the model. Each image is of size 28×28 pixels.

Dataset & Preprocessing

The MNIST Data set contains:

- 60,000 training images
- 10,000 testing images
- 10 classes (0-9)

The images were normalized by dividing the pixel values by 255.

$$\text{Normalized value} = \text{Pixel value} / 255$$

Some sample images displayed were:

5, 0, 4, 1, 9.

Results

The model achieved:

Test Accuracy = 97.49%.

The training and validation accuracy and loss were also plotted using graphs.

Five Test Predictions

Actual Predicted

7 7 2 2 1 1 0 0 4 4

All Five selected images were predicted correctly.

Dropout Experiment

As an experiment, Dropout = 0.2 was added to the model.

Dropout randomly ignores some neurons during training and can help reduce overfitting. Results:

Original model : 97.49%.

Dropout model : 97.49%.

Neural Network

The model contains:

28 x 28 image



Flatten



128 Neurons (ReLU)



10 Neurons (softmax)



Predicted Digit

The Flatten layer converts the image into one-dimensional data.

The final 10-neuron layer predicts the digit from 0 to 9.

Training

The model was compiled using:

optimizer: Adam

loss: Sparse categorical crossentropy

metric: 5

The model was trained using the MNIST training data set.

The difference b/w both models was very small.

Conclusion

The MNIST handwritten digit classification model successfully recognized handwritten digits.

The original model achieved 97.49% test accuracy, and all five selected test images were correctly predicted.

The Dropout experiment achieved 97.46% accuracy. Thus, the neural network performed well on the MNIST dataset.